



PIONEER JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION 2014

**Candidate
Name**

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**Civics
Group**

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**INDEX
NUMBER**

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H2 BIOLOGY

9648 / 03

22 Sep 2014

2 hours

READ THESE INSTRUCTIONS FIRST

Write your name, class and index number on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

**Read the instructions on the answer sheet
very carefully.**

This document consists of 2 sections:

Section A

Answer **ALL** questions.

Answers are to be written in the spaces provided.

Section B

Answer Question 5.

Write your answer on writing paper provided.

Please hand in this section separately.

Do not open this booklet until you are told to do so.

For Examiner's Use	
Section A	
1	/ 12
2	/ 14
3	/ 14
Section B	
5	/ 20
TOTAL	/ 60
4 (SPA)	/12

This document consists of **20** printed pages (inclusive of this page)

[Turn Over

Section A: Answer all questions [40 marks]

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1. Antithrombin deficiency (AD) is a blood clotting disorder in which sufferers produce insufficient amounts of antithrombin. Antithrombin binds to thrombin and herparin resulting in deactivation of the blood coagulation pathway. AD results in formation of blood clots in blood vessels which obstruct blood flow in the circulatory system.

Fig. 1.1 shows a vector map of the plasmid vector pSING003. The plasmid is used to clone the antithrombin gene, which will be inserted at the *EcoRI* site. The plasmid is subsequently introduced into *E. coli* via transformation. After transformation, the *E. coli* cells are then grown on an agar plate.

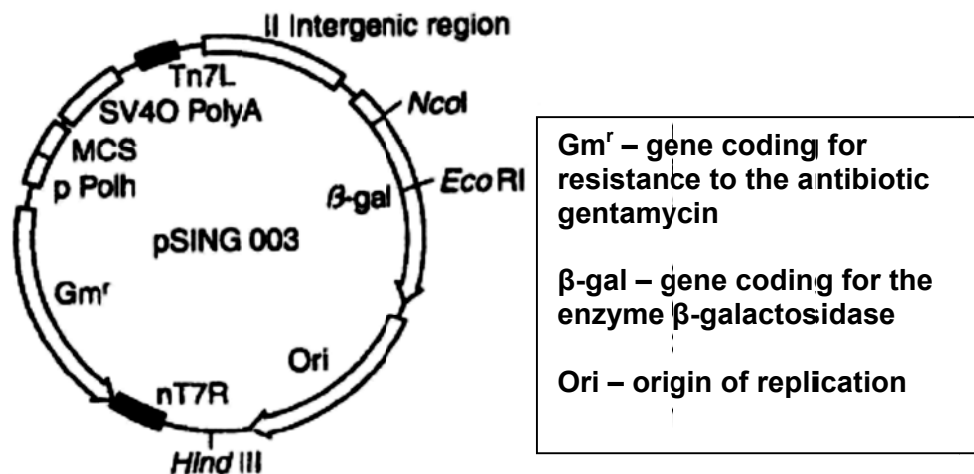


Fig. 1.1

- (a) Describe the natural function of *EcoRI* restriction enzyme. [1]

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- (b) In addition to the nutrients required for growth, name **two** other substances that are added to the agar plate to allow selection of cells successfully transformed with the recombinant plasmid. [1]

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Fig. 1.2 shows the results of plating the transformed cells onto an agar plate with the appropriate substances.

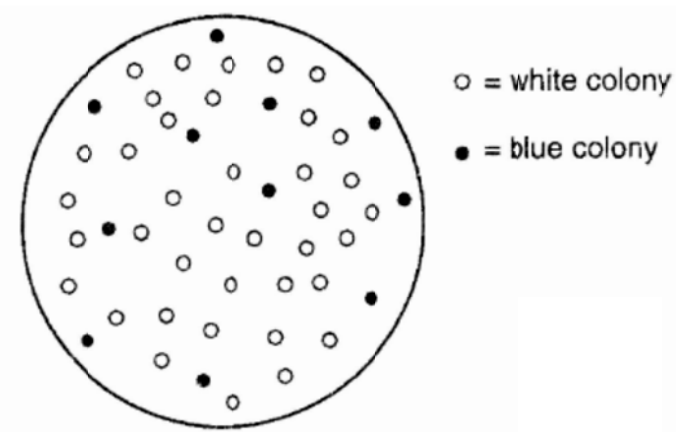


Fig. 1.2

(c) Explain how one would select the successfully transformed colonies. [2]

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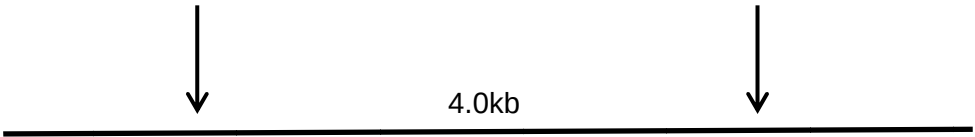
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RFLP analysis can be used as a method of detecting antithrombin deficiency (AD) in family members by using a radioactive probe to determine the presence of the mutant allele that results in AD.

Two RFLP morphs are tightly linked to the antithrombin gene which can be used in the screening of AD. Both RFLP morphs are shown in **Fig. 1.3** with arrows representing the *EcoR1* restriction sites.

Morph 1:



Morph 2:

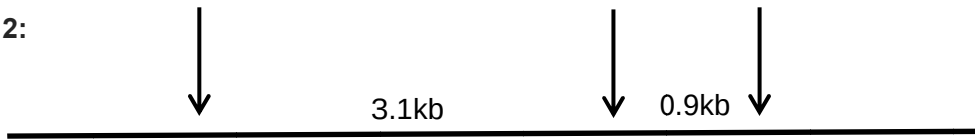


Fig. 1.3

Fig. 1.4 illustrates the Southern Blot analysis of both RFLP morphs in individuals with a family history of AD.

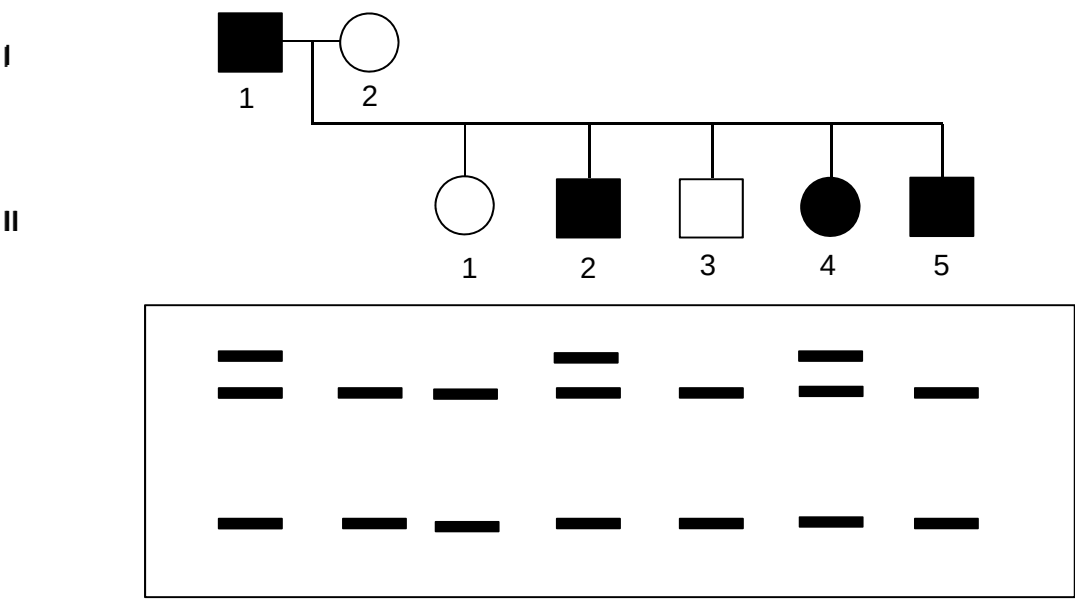


Fig. 1.4

- (d) On **Fig. 1.3**, draw the location of the probe that would produce an autoradiogram as indicated in **Fig. 1.4**. [1]
- (e) With reference to **Fig. 1.3** and **Fig. 1.4**, state and explain which RFLP morph is linked to the mutant AD gene. [3]

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- (f) Suggest the expected result of the Southern blot analysis of individual II5 and the possible reason for the difference in the DNA patterns for individual II5 from the other affected individuals? [2]

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- (g) State two features of probes that can be used to detect RFLP. [2]

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[Total: 12 marks]

2. A baby boy was diagnosed with ADA-deficient SCID at age 10 months. The boy had multiple infections, pneumonia, and persistent diarrhea and was not able to gain weight.

He received the enzyme replacement treatment for three to four months, which involved receiving injections twice a week with the necessary enzyme. However, his condition did not improve and he subsequently joined the gene therapy study in 2008. **Fig. 2.1** outlines the gene therapy process.

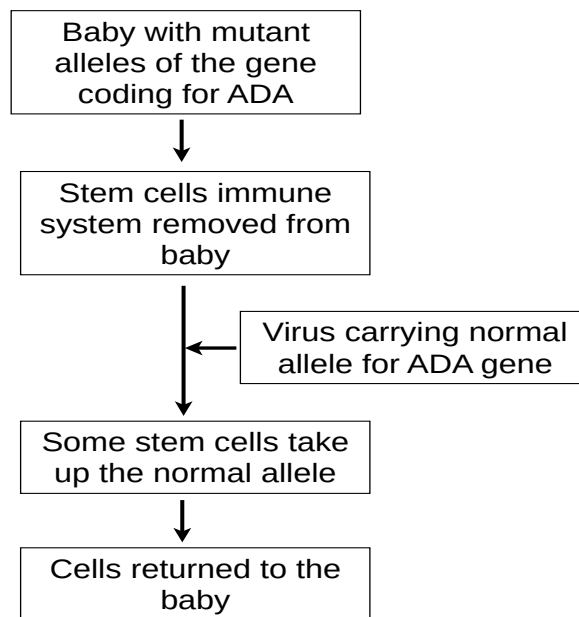


Fig. 2.1

Today, that boy, who lives with his family in Arizona, is a thriving 5-year-old.

- (a) Explain how ADA deficiency results in SCID. [3]

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(b) Suggest a disadvantage of using the enzyme therapy. [1]

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(c) Suggest the appropriate stem cells to be used in this therapy and explain why gene therapy using stem cells was more effective in treating SCID. [2]

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(d) Describe how the viral gene delivery system could have been used to insert the normal allele for ADA gene into the stem cells. [3]

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(e) Suggest two advantages of using this mode of delivery system as opposed to a non-viral gene delivery system. [2]

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- (f) The use of gene therapy to treat diseases has raised many ethical and social issues. Discuss what these issues are. [3]

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[Total: 14 marks]

3. Antithrombin factor III is a normal plasma protein (see **Fig 3.1**) that inhibits clotting by binding thrombin and clearing it from the circulation. Persons with born genetic deficiency of antithrombin factor III are at high risk to develop life-threatening clots during high risk situations such as surgery procedures due to thrombosis. Thrombosis is a condition whereby a blood clot forms within a blood vessel, obstructing flow of blood in the circulatory system. Antithrombin factor III is complex structure and often treatment doses are usually large, which posed challenges in collection in the early 1980s.

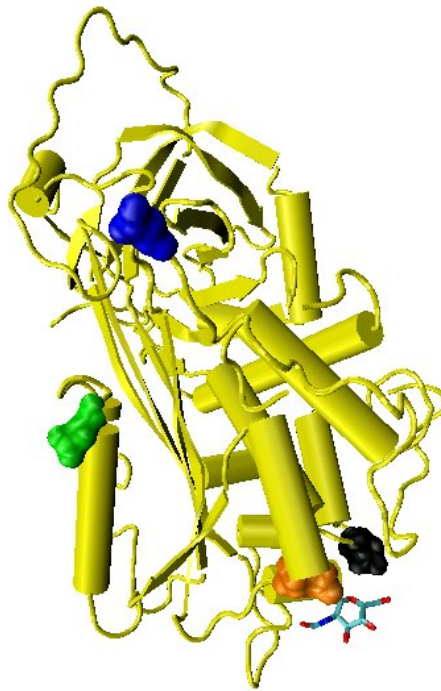


Fig 3.1

Animal pharming is a relatively new technology using whole animals to produce pharmaceutical proteins. A known example is recombinant human antithrombin factor III (rhAT). rhAT presents as an method to manage AT deficiencies by preventing abnormal clotting. A herd of dairy goats were genetically modified to express high levels of human AT in their milk to provide an abundant supply of rhAT.

Beta casein is a protein found in high concentration in goat milk. A schematic representation of the beta casein transgene for recombinant expression of rhAT is shown in **Fig 3.2**. The cDNA encoding rhAT (striped box) replaces part of the coding sequence of beta-casein gene. Restriction sites for enzymes *EcoRI*, *NotI* and *Sall* are represented by the alphabets R, N and S respectively.

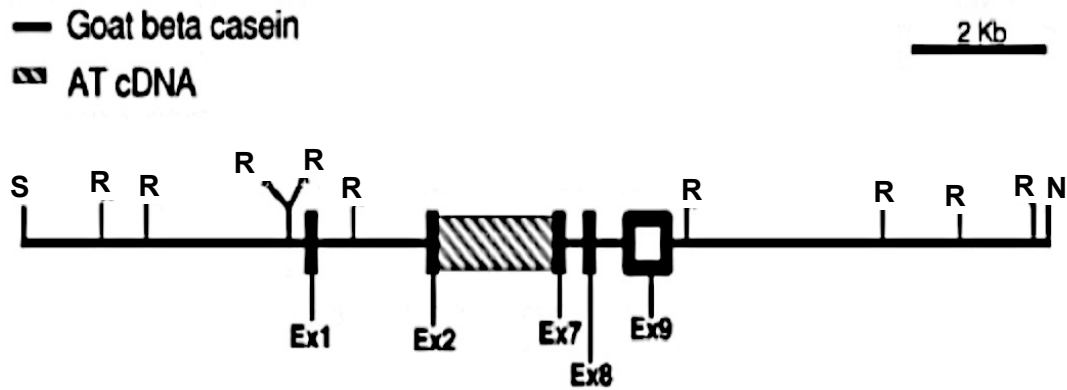


Fig. 3.2

- (a) Other than restriction sites and genetic markers, what other sequences must be present in this linearized plasmid for it to be an effective expression vector to be inserted into a mammal? Explain your answer. [2]

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- (b) Various animals were considered for the genetic modification as seen in **Table 3.3** below.

Table 3.3

Comparison of the time required to obtain recombinant proteins in different transgenic animal species

	Rabbit	Pig	Sheep	Goat	Cow
Gestation time (months)	1	4	5	5	9
Age at sexual maturity (months)	5	6	8	8	15
Time between gene transfer and first lactation (months)	7	16	18	18	33
Number of offspring	8	10	1-2	1-2	1
Annual milk yield (liters)	15	300	500	800	8000
Recombinant protein per female per year (kg)	0.02	1.5	2.5	4	40

*Gestation refers to the period of developing inside the womb

- (i) What is the advantage of using animal system instead of a bacterial platform? [2]

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- (ii) Suggest two reasons why goats are used amongst other animals for the production of rhAT. [2]

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Six surgical events on AT-deficient patients were treated with rhAT. The results were shown in **Table 3.4**.

Table 3.4

Subject no.	Age [years]	Sex	Baseline AT level [%]	Surgery type	Total rhAT received [U]	Vascular duplex ultra-sound	Clinical evidence of thrombosis	Anti rhAT antibody
1a	22	M	40	Laparoscopic splenectomy	74 480	Negative (d5)	Negative	Negative
1b	22	M	40	Bilateral hip replacement	294 000	Not tested	Negative	Negative
2	47	F	49	Hysterectomy	101 921	Negative (d7)	Negative	Negative
3	36	F	58	C-section	65 000	Negative (w6)	Negative	Negative
4	72	M	52	Coronary artery bypass	39 200	Not tested	Negative	Not tested
5	71	M	53	Knee replacement	73 480	Negative	Negative	Negative

- (c) Comment on the effectiveness of rhAT produced and refined from transgenic goats. [3]

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- (d) Besides animal pharming, animals have also been genetically modified to increase food yield. A well-known example would be AquAdvantage salmon. AquAdvantage salmon is a transgenic salmon where a promoter for the anti-freeze gene of an ocean pout and a gene for the Chinook growth hormone was inserted into a plasmid pUC18. The recombinant DNA (**Fig. 3.5**) is then introduced into fertilized egg cells of the Atlantic salmon by microinjection and stimulated to become triploid.

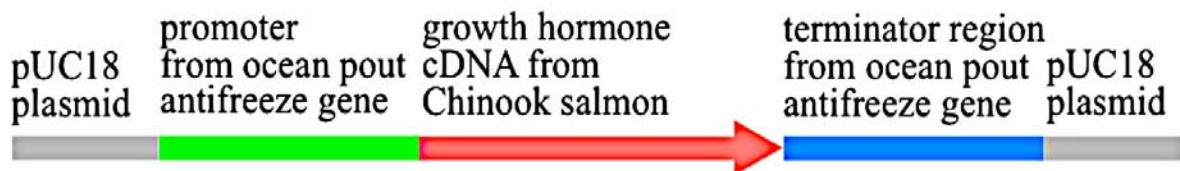


Fig. 3.5

- (i) Explain why the promoter from ocean pout antifreeze gene is used. [1]

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(ii) The egg cells were induced to be triploid. Explain the significance of this. [2]

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(iii) Describe one ethical and one social concern that might arise from the production of AquAdvantage salmon. [2]

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[Total: 14 marks]

4. Planning Question [12 marks]

A new breed of carrots was tested for sucrose concentration. Freshly squeezed extract was obtained for this investigation.

Your planning must be based on the assumption that you have also been provided with the following materials which you must use:

- Solution of unknown concentration of sucrose extracted from the new breed of carrots, labeled P
- 10% sucrose solution, labeled S
- Distilled water
- Dilute hydrochloric acid
- Sodium hydrogen carbonate powder
- Benedict's solution
- Colorimeter and 1.5 cm³ cuvettes

A colorimeter is a device that can be used to measure the concentration of a solute by measuring the absorbance of different wavelengths of light in a solution. Generally, a solution of higher concentration will absorb more light energy at a specific wavelength. Therefore, the higher the absorbance value, the higher the concentration of a solution. Although absorbance has no units, it is often recorded as absorbance units (A.U.).

You may select from the following apparatus and use appropriate additional apparatus:

- normal laboratory glassware e.g. test-tubes, beakers, measuring cylinders, graduated pipettes, glass rods etc
- syringes
- white tile
- Bunsen burner with tripod, gauze and bench mat
- timer e.g. stopwatch or stop clock
- thermometer

Your plan should

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it,
- an explanation of the theory to support your practical procedure,
- identify the independent and dependent variables,
- describe the method with scientific reasoning used to decide the method so that the results are as accurate and reliable as possible,
- show how you will record the results and the proposed layout of results tables and graph with clear headings and labels ,
- include reference to safety measures to minimize any risks associated with the proposed experiment,
- use the correct technical and scientific terms.

[Total: 12 marks]

[illegible]

This image shows a full page of white paper with horizontal dotted lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting practice. There are no margins, text, or other markings on the page.

Section B: Essay Question [20 marks]

Write your answers to this question on the separate answer paper provided.

Your answers:

- should be illustrated by large, clearly labelled diagrams, where appropriate,
- must be in continuous prose, where appropriate.
- must be set out in section (a), (b), etc., as indicated in the question.

5 (a) Outline, using a named example, how the micropropagation technique is used in production and cloning of pest resistant plants. [6]

(b) Describe how RFLP analysis is used to support the process of linkage mapping. [6]

(c) Describe, using a named example, how RFLP enables disease detection. [8]

The End